

Dr. Kevin Mugoye14:23:59

It is going to be very interesting to see how you are going now to align it.

to align it now to switch information system.

Because one of the questions that you're going to be asked is, what is your contribution to the body of knowledge in information systems?

That is one of the questions that I'm going to ask, and somebody else is going to ask you, but for the... with respect to the title and topic, it's a very good topic, a very good title, at the... even at the level of PhD.

That one, I'm happy about the title.

The only problem I will have.

He's coming to the implementations on what you want to achieve and then how you are going to fit it into your field of study, which is information systems. Okay.

But, uh, that is something that you can be guided, and we see how far it goes.

Yeah, why is that important, Chair? That is very important because.

The output that is anticipated at each of these levels which I've said is different.

Okay? The, I can say, the Riga.

Outputs. When you take this title to cybersecurity, then the output is almost anticipated.

When you take it to computer science, the output is almost anticipated. So then, with respect to IS, maybe, I don't know whether it will be very unfair to judge it based on the previous two, because it is a...

it is something that you should think about how to scale it down, so that it doesn't give you enough... a lot of problems, huh? Going forward. Okay, it... then...

While I thank you about the title.

Then I have now my question would be.

My question would be one.

All right.

What is your contribution?

To the body of knowledge.

in the space of information systems. What is the major contribution that is supposed to earn you.

PhD in the discipline of IS. Wow.

You can work with that first.

Martin Muriki14:26:17

Yes, yes, thank you.

Um...

Yes, I accept that my topic more or less falls into.

information into cybersecurity, more or less, as you've stated.

And whereas I'm doing an information system, but for my case, I hope to achieve.

to, to have a, a novel blockchain enabled multimodal system. So I'm going to like have a framework that is going to, to come up with a new framework that is, that is going to solve a.

Uh, the issues that, uh, that are already there.

in, uh, in, in, we expect, we are experiencing at, at KCA.

So I'm going to contribute to information systems theory by extending the application of ISDT.

and the design science, uh, research. So I want... I hope... I hope to contribute into this ISDT theory. So that is what I had in mind, uh, Dr. Lee.

Dr. Kevin Mugoye14:27:27

Yeah, very well answered. I like that. You talk about a framework.

So then if that is the case, you should include the word framework in the title, which now clones it down. We don't have to expect so much because if it goes the way it is, then there will be a lot that will be expected.

Going forward, yeah?

Martin Muriki14:27:49

From...

Ya no.

Dr. Kevin Mugoye14:27:51

which might be misleading to the audience. Then now let's come to the main objective.

Martin Muriki14:27:54

No te.

Dr. Kevin Mugoye14:28:08

I was listening at your answer in terms of contribution. You're going to, I will ask you at your free time. to read about a Phd. Contribution.

What qualifies to be a contribution?

in a PhD.

And what does not qualify to be a contribution for PhD in academic sphere?

And why I'm saying this, I'm saying this intentionally because one, you've said you're going to extend the application of something.

Then the question would be, by extending an application of something, do you think that is sufficient to earn you a PhD?

Martin Muriki14:28:42

Y así.

Dr. Kevin Mugoye14:28:57

Do you think...

Just extending the application of something.

would be strong enough to earn you a doctorate.

That's my question.

Martin Muriki14:29:14

Tu cama, puedes.

Norte, aparte de standing them.

Tenía un.

Arquitectos vendiendo discos, projects.

Dr. Kevin Mugoye14:29:32

Okay, that is beautiful. Now that will be your contribution, which now forces you to go back to the drawing board and reframe your title.

Reframe your problem, because what needs to come out clear is your contribution.

Okay, that is the main piece of work which you are going to see. You expect to see in this academic journey.

So then my suggestion would be for you to be safe. You know, a PhD is viewed by the global society, not a local society at master's level.

Martin Muriki14:30:01

Mhm.

Y este.

Y es.

Dr. Kevin Mugoye14:30:08

The institution of learning is given the mandate.

to reward and award masters, right? But when you could look at PhD, you are pushing yourself to the global community.

Martin Muriki14:30:15

Yes.

Dr. Kevin Mugoye14:30:20

And the danger is anyone can criticize your work, right?

Martin Muriki14:30:25

Bien, no.

Dr. Kevin Mugoye14:30:26

Yes. So then, to be safe, what I would want, I would request that you do.

That that you've explained right now should come out as the contribution which you're going to do.

Martin Muriki14:30:35

Mhm.

Dr. Kevin Mugoye14:30:40

in this study. And that contribution is novel because in a PhD, your contribution should be a novel, should be novel, right?

Martin Muriki14:30:47

Tus 10.

Dr. Kevin Mugoye14:30:49

Yes. So that is what I feel should be the subject under discussion.

which would form even your problem and your objective. Then if you do that, I think I'm good. To me, I'm okay.

But if you let it be the way it is, you will be inviting a lot of problems.

Going forward. Okay.

Martin Muriki14:31:12

Yes no tan destacable.

Dr. Kevin Mugoye14:31:12

Now.

Okay, thank you very much. Now.

I will not have a problem at the moment. I'll let others engage on the objectives, on the specific objectives.

But I would want to ask another question on that.

I have a question.

On the theory.

You mentioned something to do with the seer triad.

And you cited a researcher. Can you go back there? You cited a researcher.

On the theories, under theories, you did mention.

Not good. No.

Martin Muriki14:31:51

Esta minutes.

Dr. Kevin Mugoye14:31:55

Yeah, there, I think there. Yes, you mentioned about the shared theory.

And you cited a researcher.

whose work was done in 2021.

I want you to explain to this panel.

OK, what you mean?

When you talk about the theory, and then you reference that particular researcher.
does it mean that the seer triad theory, or what is seer triad theory, according to you? And when did it begin? So that we can know, what did this white man, Admatphon, do?

Would that in that respect?

So just answer on this hand first.

Martin Muriki14:32:37

No sur.

Social, transformerties confidentiality Integrity anda Verhability Security Free.

Yetis Yace decía.

Ya ya ya.

Todo el mundo.

Desde el punto de.

Sobre de inscription the access controles.

So this one ensure that the, the, the biometrics are secure in nature and then for the integrity.

we don't want somebody to come and change the exams records. At the end of the day, once the students have been able to.

To submit the exams, you want them to be what you call tamper-proof.

This is going to be achieved by what we call the hashing.

And then availability. Yes.

Dr. Kevin Mugoye14:33:53

Yeah, that is okay.

That is okay. My question was this.

When was the SIA triad?

I don't know what I don't want to say discovered.

Okay, what does the original seer, when was it discovered?

Then what did this researcher do in 2011, the one you've cited?

With that respect, because, yeah, what I'm trying to say is that the SIA triad existed even before 2011, right?

Martin Muriki14:34:26

Mm-hmm. Yes.

Yes, yes.

Dr. Kevin Mugoye14:34:29

And when you make such a citation, it means.

There is an original meaning to the CF framework or model with respect to information security. But then there was a variation.

That was introduced by this researcher in 2021. And your study, you then want to adopt this variation, not the original. That is what it might imply here. So the question I'm asking is, does it mean whatever I have said?

Does it mean that this white man and Matt Ford changed the seer?

That is already, by default, we know.

Martin Muriki14:35:13

Yes, the original CIA was back then in the 1970s.

Uh, but, uh, the Whitman, Whitman, uh, I think with Akina Mako, they now actually, uh.

Uh, formulized it and talked it into in details.

And one of the things, the main things they introduced in this, um...

Uh, in this... in this study of 2021.

Uh, was to...

was to, was to, was, was to, to, to, to take it to another level from the, to improve the actual, the initial model.

Dr. Kevin Mugoye14:35:50

Let me stop you.

Martin Muriki14:35:52

That was there in the 1970s. So, and it is just like, it was a very, it was a foundational framework.

Dr. Kevin Mugoye14:35:53

Yes.

Martin Muriki14:36:00

that they used for these information securities.

So they improved the model further.

Dr. Kevin Mugoye14:36:06

Okay.

So, what you do...

What you do then, it is fine, but you need to be very careful when you're making such a...

That's a decent answer.

Because it will not harm you if you just maintain the seer.

And speak about what is it that you now want to adopt?

But if there is something, there's a revision that was done, and it's different from the original SEER, then you're allowed to go this direction, and then mention what is it.

Okay. That now changed. And then you are safe. And that applies. That should apply for all the theories that you've cited here.

Okay.

If, in case.

That was not the year in which the original version.

or that particular concept or theory was reformulated.

So that is the notion that you can adopt, and it will save you a lot of trouble going forward.

Okay, then.

I saw something like a conceptual framework. Do you have a conceptual framework?

Of what you want to do?

Somewhere.

Martin Muriki14:37:17

Yes, I have, but I have not put the, the actual picture of the same, but I have it in my paper. I can share the paper. I have it open. It's, it's somewhere here.

Dr. Kevin Mugoye14:37:23

Okay.

Okay, yeah, the the conceptual framework. The conceptual framework is ideal.

Martin Muriki14:37:29

Yes.

Dr. Kevin Mugoye14:37:32

Because it now shows, it communicates to us.

the static structure of what we expect in the whole problem domain.

Okay.

Martin Muriki14:37:43

Yes.

Dr. Kevin Mugoye14:37:43

So here is what you're going to speak to us by you telling us this is what I'm going to, I'm introducing, this is what is going to bring the effect that I want, and this is the final output that we're going to see and do all of our things. So what I would advise you to do again.

Martin Muriki14:38:08

Mm-hmm.

Dr. Kevin Mugoye14:38:12

Okay.

Martin Muriki14:38:14

Yes, ma'am.

Dr. Kevin Mugoye14:38:22

Independent.

W.

Chu chu chu get from This.

Of the outputs best on.

Martin Muriki14:38:45

I...

Dr. Kevin Mugoye14:38:53

M.

Martin Muriki14:39:07

I was trying to simplify it, actually, for easy understanding.

Yes.

Dr. Kevin Mugoye14:39:43

Okay, so that the conceptual framework does not need to be misleading.

And in our case, based on our problem, however, I've understood it.

Based on the problem which you've shared.

You want to come with a multimodal blockchain enabled.

Right?

Martin Muriki14:40:04

Mm-hmm.

Dr. Kevin Mugoye14:40:04

authentication model for improved security. For improved what? Security.

Martin Muriki14:40:11

Yes.

Dr. Kevin Mugoye14:40:11

So...

We want to see this multimodal authentication model. We want to see it. That is the... that is the framework that we want to see, and the conceptual framework. How will... how will it look like?

Then how is it going to guarantee an improvement with respect to security?

Based on what the title is speaking, that will be a major factor that we'll be seeing, we'll be looking for in the conceptual framework.

Okay.

Martin Muriki14:40:41

Yes, yes, that's right.

Dr. Kevin Mugoye14:40:43

Otherwise, it's a very good piece of work that if enhanced, it can go places. I want to stop there so that I don't sound very boring.

Back to you, Chair, and thank you very much.

Dr. John Wanyoike14:40:58

Okay, thank you very much, uh, Dr. Kevin Murray, thank you very much, uh, Matthew Morik.

It is now time to have comments and also to receive comments and also questions.

from Dr. Rufus Gekeira. Dr. Rufus Gekeira, he is raising up his hand.

Okay.

Dr. Gikera14:41:15

Yeah, thank you so much.

Dr. John Wanyoike14:41:15

So. Yeah.

Dr. Gikera14:41:21

I can proceed.

Dr. John Wanyoike14:41:24

Yes, you can proceed, Dr. Rufus.

Dr. Gikera14:41:28

Yeah, thank you so much for this opportunity, Chair.

And I also want to, uh...

Congratulate Wanamati Muriki for...

A very good presentation.

Uh, having reached this milestone is no mean feat.

I really want to just to encourage you to move on.

Uh, maybe my question, and of course your slides are well done, maybe my question, um.

Uh, if you don't have an answer, you can just note it down.

Because most of the comments, I'll share them on the document.

But we find most of the security solutions in the e-examination systems.

Most of them, they fail because their authentication is weak.

Or because maybe of the human procedural kind of issues.

Maybe you can answer that.

But the main, for me, is the fact that...

most of the blockchain systems, they are criticized because of their latency. You know, the time duration between when you make a request and you get a response.

And, uh, maybe... eh... what...

the strategy have you put in place to ensure that you get kind of an optimized kind of a performance? Considering that, you know.

um...

There are a number of users at any particular time.

But other than that, your work is good and I wish you all the best. Thank you.

Martin Muriki14:43:19

Thank you, Dr. Lee, for that good question, thank you. Uh, now, uh, yes, your point is valid, and what you've said.

Biometric systems, by the way, they are a bit heavy and they take time to...

to... yes, I just said about lagging. Now, I was thinking of incorporating edge computing.

at...

So, that is what I had in mind, you kindly correct me if I'm wrong, but I was thinking of using... incorporating edge computing, whereby most of these, uh, processes, they don't have to go the way to the server. Uh, then they come back to the user. So, if most of these processes, once the student has logged in for the first time.

Then, most of those other processing can be done locally.

Uh, instead of each time, uh, communicating with the server back and forth, like that.

That is what I had in mind. Don, what's your take, Dr. Ali?

Dr. Gikera14:44:20

Yeah, to me, edgy would be a good direction to explore. I'll give the other comments through the documents. Thank you.

Martin Muriki14:44:28

Thank you.

Dr. John Wanyoike14:44:29

Okay, thank you very much, Dr. Rufus Gekera. Thank you very much, Martin Moriki. It is now time to receive comments and also questions from Dr. Richo Kebuko. Dr. Richo Kebuko, it is now your turn.

Dr. Kibuku Rachael14:44:44

Thank you. Thank you, Chair. And thank you, Martin Ulicki, for your presentation.

However, I came late because I was in another meeting. I was trying to combine, uh, two meetings using the same license, and I was being kicked out from this one.

However, I found you trying to argue out a point which is okay, that's the business of the PhD.

Oh...

I want to comment.

I just want to make a suggestion, uh, where you're talking about TAM and bootcuts.

It's a suggestion, you just go think about it, and then you can see... you and your supervisors can see how best to deal with it.

Utwat, the 2003 that you have, uh, presented, or you have reviewed.

Uh, treated by vacuum test.

was updated 10 years later, so we have Oathwatch 2.

And UTWAS itself is a combination of eight theories, CAM included.

So I, I feel that the papas.

term would serve is the same purpose, or would be served better by outwards.

Uh, because then when you have both of them, it is repetitive.

And therefore I would suggest it's a suggestion because term is already included in what you drop the term. and deal with OTWAT.

And, uh, maybe in your review, uh, you could be telling us what... because, uh, Vacante Chatel, uh, a decade later, they came up with Wootwatch 2.

It would also be good to look at other people who have improved, or modified, or extended, or whatever they have done when reviewing, and please make sure that it relates to your work.

Of course, that is the part that Dr. Magoy was asking you, the relationship with information systems, make sure that it comes out very clearly because the PhD we will award you will be titled PhD in information systems, and they will ask you what is your contribution in.

Uh, that field, or discipline, or information system. So, that is one of the theories that adapts these systems that you're talking about, even the one that you will create, uh, to improve examination security.

Then I also want you to, uh... when you're reviewing the literature, we are happy that you're still at the proposal. So, you know, literature review is a close life cycle activity throughout this PhD.

So we would add you to figure out the theoretical contribution that you shall make or that this PhD will make.

Of course, I see you propose to develop a model.

And we anticipate there will be some good results. We are eager to see those results when you return with the thesis.

But you would want to see, probably, what gaps will you have identified in these theories that you've reviewed, and how you will propose to fill those gaps, or what will be your theoretical contribution.

Thank you so much, Martin, for doing what you're doing. I am very proud of you. Thank you.

Dr. John Wanyoike14:47:59

Okay, thank you. Okay, thank you very much, Dr. Ali Tokebuku. Thank you very much, Martin.

Martin Muriki14:47:59

Thank you, doctor.

Dr. John Wanyoike14:48:06

Okay, Martin, you have to... are you going to...

Answer anything or I just proceed.

Martin Muriki14:48:14

Thank you, Dr. Kibuko. Thank you. I'll make the changes you've recommended. I'm sorry. Even my supervisor had mentioned that I do away with the term.

I'll do that. Thank you for that. I'll do that. Thank you. Thank you, Dr. Ali.

Dr. John Wanyoike14:48:31

Okay. Thank you very much, Martin. So it is now time to receive comments and also questions from Professor Lucy.

Professor Lucewar Uguru, it is now your turn.

Prof. Dr. Lucy Waruguru14:48:46

Yeah, thank you, Dr. Inouye, and thanks, Martin.

Um, I want to commend your work, just like the others, especially the solution that you're coming up with.

Uh, because you are trying to integrate the multimodal biometrics with blockchain, and also focusing on exam, uh, electronic exams.

Which is, uh, something that...

most studies have not done. They've probably done one or the other, but not, uh.

All the three together. So that is commendable.

Um, I had, uh, many questions, but I think in the interest of time, because a lot has already been said, I will try to avoid repetition. I will just focus on some recommendations and maybe one or two questions.

Uh, the first, maybe we just look at your title. I thought something is missing from your title. It was not...

Complete to me.

You say blockchain enabled multimodal biometric.

Martin Muriki14:49:48

Authentication, sorry, sorry. Authentication, vermetic authentication.

Prof. Dr. Lucy Waruguru14:49:49

Do you... is it?

is...

Ah, okay, okay, because I thought something was missing there. I didn't know whether it was system, authentication, or framework, or whatever it was, yeah?

But then, I have also several, uh...

recommendations. The first one is to do with your... what I saw in your background, and also problem statement, both in your presentation and also in the document that, uh...

you had submitted. Um, it is very important that when you prepare the background.

And also the problem statement, you need to contextualize.

If you don't contextualize, then most of the...

most of the, the sentences there become very generic, yeah? So you are provided some statistics of KCA University, but I believe that you, there are a lot of, uh,

other contexts that you could provide, maybe within Kenya, within Africa, and so on, so that it becomes more... more of a contextual issue, because otherwise, then you find that this... I think there was a statement you made where you say that the African.

Important.

Nordiskehjælp.

Den sekund.

Is also recommendation and It is with respective objectives.

Maybe you can just go to the objectives. I have two questions there. So the first would be to strongly recommend that you don't...

I mean, for your PhD study, you don't restrict yourself to the model, to the authentication model, but instead, maybe you can try to adopt a framework. You can build a whole framework that incorporates.

models, yeah? We're expecting that you're going to publish several articles, and each of these will be independent research. So, if you have a framework, it kind of helps to encompass.

Everything else that will be within that umbrella, that could be the different models. So maybe you could change this to maybe a blockchain enabled multimodal biometric authentication framework for higher education institutions.

Uh, rather than...

Uh, multimodal.

uh, biometric authentication model, yeah? So, uh, that is another, a second recommendation. And then the third is with respect to your context, uh, your conceptual framework.

Uh, just quickly go to the framework. I know you said that a more comprehensive one is in the... document, but this one kind of tries to capture quite a lot, and we can discuss based on it.

Uh, so yeah, the first I did not understand is how you have, uh, operationalized these variables. Like, for example, you've said facial recognition.

fingerprint recognition, voice recognition, biometric vision. I did not understand how, what exactly are we expecting?

as independent variables there, yeah?

And then.

Uh, you have also provided a dependent variable that you have called improved E exam.

System security. I had two issues with that one.

The first is that variables usually should not be suggestive. You can't say improved because you are not sure that it will improve. It can actually deteriorate. So variables usually have values that go up or down.

as based on the input that you provide, yeah, as the independent variables. So, um, I would suggest here that at you somehow make your variables neutral, but at the same time.

If it's important to clearly operationalize, like for example, when you say E examination system security.

Uh, you know, information security is very broad.

So that here we are talking about, I mean, thinking about it broadly, I'm suspecting that it will probably incorporate.

Martin Muriki14:54:24

Yes.

Prof. Dr. Lucy Waruguru14:54:32

对啊。

The hardware security. Maybe protocols, they are goinges. I don't know which specific one. Are you intending to look at all of them in entirety? Or are you looking at a specific.

A specific perspective that you're going to focus on. I think it would be important, and i believe the operationalization we should have brought that out.

So um.

I hope that one is the document, but i think it would have been good also talk about it.

Uh, and in relation to that, now to my last, uh, also question.

How are these variables?

Related to the research instrument that you presented in the methodology.

Because when you presented the research instruments, I did not see any instrument that is going to help to capture what you can use to construct your independent variables.

You talked about a survey study.

I mean, I think you talked about questionnaires, interviews.

Yeah, maybe you can just go there.

Yeah, here, you talked about the questionnaires, interviews, uh, observation checklists...

And, uh, prototype evaluation. So maybe you can explain as the last question, and then I'll take it back to the chair.

How are these data collection instruments building up to your conceptual framework?

Martin Muriki14:56:18

Thank you, Prof.

Um...

Like, uh, I would use, uh... my system is going to have, uh, facial recognition, fingerprints, voice... and then the blockchain technology. Now, for when I'm going to do my data collection, I was thinking of developing a prototype that is going to.

To enable me to capture at least some, uh...

some data from the users. From this data, I'm going to...

You missed out?

System.

Uh, that's what I demand, uh, prof.

Thank you.

Dr.john Wanyoike14:57:32

Okay, thank you. So, Professor, are you, you have the room?

Eh, Professor.

OK, thank you. So thank you very much, Professor. Thank you very much, Martin Moriki. So Dr. Buenjin a, Dr. Omori, you cannot ask any question to this student because you are supervisors. So Mr. Martin Moriki, I'm going to ask you just a very general question.

How will you be able to ensure that the extra security provided by using blockchain and also several biometric methods?

is going to be worth the extra money, the extra effort, the extra time needed to run the system.

Martin Muriki14:58:18

Thank you, Dr. Salim.

Uh, we, we've seen that, uh, that we have had the issues of impersonation in, uh, during exams.

And, uh...

At times, if the number is big, and students realize that they can not study for the exam, then look for some body who is bright and then ask that person to do the exams on their behalf.

Then if that becomes a norm and then it many students start practicing it, it may become an issue because I and I don't want to mention schools, but you had a school in Nairobi, a technical school at. was issuing certificates to students without them going through the due process, the academic process. The institution was closed down.

So, I think the initial capital for the infrastructure might be a bit high, but at the end of the day, it's going to , for our reputation of the school, uh, I think it's going to be worthwhile.

And, um, it's going to avoid so many issues, like, uh, the cheating. Students will know that they need to have, uh, to... to study for them to... to do the exams. There's... there's no shortcuts.

So, my system might be slightly a bit expensive, but...

At the end of... at the end of the... at the end of the day, for... for compliance purposes, we'll be sure that we are issuing genuine certificates to genuine students who have attained a...

who have learned something in school. Thank you, Dr. Ellie.

Dr.john Wanyoike15:00:00

Okay. Thank you. Thank you very much, Martin Moriki. Let us have the next presenter. So the next presenter is Nicodemus Muabera.

Are you ready?

Nicodemus Mwambela15:00:13

Yes, I'm ready.

Dr.john Wanyoike15:00:14

Thank you very much. You can open your camera so that we can be able to see you.

Then you close your camera and start the presentations.

Yes, thank you very much. We have been able to see you start there for now the presentations.